Stanley Nelavala

Software Engineer: Backend and Infrastructure

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Huntington, IN | Open to relocate anywhere

Summary

Backend engineer who measures before and after. On a 328,000-device dev database I cut a deep catalog query from 285 ms to 0.20 ms, designed the signed service-to-service API that every platform module uses, and fixed a regular expression denial-of-service bug. Public Python, Rust and Go services with tests, traces, load tests and failure labs. M.S. in Computer Science (4.0 GPA). AWS Certified Solutions Architect, Associate.

Experience

Software Development Engineer Nov 2025 – Present
Mass Technologies, team of 10 (RxApps360 hospital platform, pre-launch; numbers from dev and test) *Huntington, IN*

- Replaced offset paging with keyset paging on the catalog device list: on a 328,000-device dev database, the query for row 150,000 fell from 285 ms to 0.20 ms. Search pages never skip or repeat rows, tested with rows inserted, deleted and tied between pages.
- Cut the planning Items page from 3 API calls and 162 KB to 2 calls and 37 KB with typeahead search and cursor paging.
- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.
- Fixed a regular expression denial-of-service bug in a shared validation package: an 80-character hostile input took about 138 s; the new linear pattern matched the old results on 60,000 random inputs.
- Built RxDesign, the hospital design service, backend and web (FastAPI, React, TypeScript), with a deterministic, rule-based layout engine that places rooms on a 0.5 ft grid with doors, exits and flow lines; 220+ offline tests on the API.
- Built the RxAsset iOS and Android app (React Native, Expo) with scan-to-search and an on-device voice assistant (no cloud call).

Adjunct Faculty (part-time) Jan 2026 – Aug 2026
The College of St. Scholastica *Duluth, MN*

- Taught Programming II (Java) and Machine Learning for Data Science.

Graduate Research Assistant and Teaching Assistant Aug 2023 – Aug 2025
University of Minnesota Duluth *Duluth, MN*

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR) that collected and analyzed 1M+ TikTok videos from 300k+ authors for thesis research on misinformation, then grouped themes with BERTopic; presented at the UMD CS graduate poster session. Led labs and office hours for Python, NLP and Parallel Programming.

Projects (public code, tests)

Webhook and Connector Gateway | *Rust, axum, tokio, PostgreSQL, SQLite, sqlx, OpenTelemetry, k6* GitHub

- Rust gateway: HMAC-SHA256 webhooks verified in constant time, per-tenant rate limits and idempotency, durable store before delivery, retries with backoff, dead letters with replay and circuit breakers.
- Global backpressure (503 with Retry-After) and a kill -9 failure lab where 500 of 500 events reached the downstream once after redrive; delivery SLO alerts tested with promtool.

Inference Gateway | *Go, gRPC, ONNX Runtime, PyTorch, OpenTelemetry, Prometheus, Helm* GitHub

- Go service for an int8-quantized banking-intent classifier over HTTP and gRPC with micro-batching, deadlines and load shedding; macro-F1 0.8636 for both fp32 and int8 on the BANKING77 test split.
- One-factor experiments and pprof ruled out GC and tokenizer cost; a Little's law model sizes a 10x launch peak at 2 to 3 replicas (local benchmark).

Agent Integration Platform | *Python, FastAPI, LangGraph, MCP, PostgreSQL (pgvector), Redis, Terraform* GitHub

- LangGraph case workflow with typed state, a Postgres checkpointer, human approval that resumes after restart, and step and token budgets; 62 deterministic replay evals, prompt-injection tests, PII redaction and an append-only audit log.

Skills

Languages: Python, Rust, Go, TypeScript, JavaScript, SQL, Java
Backend and infrastructure: FastAPI, axum and tokio, gRPC, REST, PostgreSQL, Redis, Kafka API, idempotency, backpressure, Docker, Kubernetes (kind), AWS
Performance: pprof, k6, Locust, OpenTelemetry, Jaeger, Prometheus, query plans and keyset paging, capacity models
AI and ML: PyTorch, ONNX Runtime, int8 quantization, LangGraph, MCP, deterministic LLM evals; AI-assisted development (Claude Code, Codex, opencode, DeepSeek)
Frontend and testing: React, React Native, Playwright, pytest, property tests

Education and Certification

M.S. Computer Science, University of Minnesota Duluth, 2023 – 2025, GPA 4.0, Graduate Summer Research Fellowship 2024 and 2025
B.Tech Computer Science and Engineering, IIIT D&M Kancheepuram, 2019 – 2023
AWS Certified Solutions Architect, Associate, Feb 2025 – Feb 2028 (verifiable on Credly)